

3a) left off at

$$I_3 = \int \frac{z^{1/2}}{(z - \frac{1}{\sqrt{2}}(-1+i))(z - \frac{1}{\sqrt{2}}(-1-i))} dz$$

in my extra notes I wrote:

$$\cancel{e^{3\pi i/4} = -1+i} \quad \text{and} \quad \cancel{e^{-5\pi i/4} = -1-i}$$

But this should instead be

$$\cancel{*} \quad \boxed{e^{3\pi i/4} = -\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}} \quad \text{and} \quad \boxed{e^{-3\pi i/4} = -\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}} \quad \cancel{*}$$

→ Sorry about this

Thus,

$$I_3 = \int \frac{z^{1/2}}{(z - e^{3\pi i/4})(z - e^{-3\pi i/4})} dz$$

Using Cauchy's Res. Thm:

$$I_3 = 2\pi i \sum \text{Res}(I_3);$$

$$= 2\pi i \left[\left. \frac{z^{1/2}}{(z - e^{3\pi i/4})} \right|_{z=e^{-3\pi i/4}} + \left. \frac{z^{1/2}}{(z - e^{-3\pi i/4})} \right|_{z=e^{3\pi i/4}} \right]$$

$$= 2\pi i \left[\frac{e^{-3\pi i/8}}{e^{-3\pi i/4} - e^{3\pi i/4}} + \frac{e^{i\pi/8}}{(e^{3\pi i/4} - e^{-3\pi i/4})} \right]$$

$$= \frac{2\pi i}{(e^{3\pi i/4} - e^{-3\pi i/4})} (e^{3\pi i/8} - e^{-3\pi i/8})$$

Note, $e^{-3\pi i/4} = e^{5\pi i/4}$ so

$$\begin{aligned} e^{3\pi i/4} - e^{-3\pi i/4} &= \left(-\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right) \\ &= \frac{2}{\sqrt{2}}i = \sqrt{2}i \end{aligned}$$

$$\Rightarrow \frac{2\pi i}{(\sqrt{2}i)} (e^{3\pi i/8} - e^{-3\pi i/8})$$

Since $\sinh(x) = \frac{e^x - e^{-x}}{2}$

$$= \frac{3/2 \cdot 2\pi i}{\sqrt{2}i} \left[\frac{e^{3\pi i/8} - e^{-3\pi i/8}}{2} \right] = \underline{\underline{2^{3/2} \pi i \sinh\left(\frac{3\pi}{8}\right)}}$$

